

习题六

1.11) - BTJ 12) - BTJ

$$2.11) y = 5x^2 \quad y' = 10x \quad x \cdot 10x = 10x^2 = 2y \Rightarrow \frac{12}{16}$$

$$12) y = \sin(2x+C) \quad y' = 2\cos(2x+C) \quad \left(\frac{dy}{dx}\right)^2 + 4y^2 - 4 = 4\sin^2(2x+C) + 4\cos^2(2x+C) - 4 = 0$$

$\Rightarrow \frac{12}{16}$

$$13) 1^\circ y_1 = xe^{-2x} \quad y_1' = e^{-2x} - 2xe^{-2x}, \quad y_1'' = -2e^{-2x} - 2(e^{-2x} - 2xe^{-2x}) = 4xe^{-2x} - 4e^{-2x}$$

$$y'' + 4y' + 4y = 0 \Rightarrow \frac{12}{16}$$

$$2^\circ y_2 = e^x \quad y_2' = y_2'' = e^x$$

$$\therefore y' + 4y' + 4y \neq 0 \Rightarrow \frac{12}{16}$$

$$14) y' = \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = -e^{-t} \cdot \frac{1}{(t+1)e^t} = \frac{-1}{(t+1)e^{2t}}$$

$$(1+x)y' + y' = (1+t) \cdot \frac{-1}{(t+1)e^{2t}} + e^{-2t} = 0 \Rightarrow \frac{12}{16}$$

$$3. y'(0) = y(0) = 0, \quad y''(0) = 1$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{f(x)} = \lim_{x \rightarrow 0} \frac{\frac{2x}{x^2+1}}{f'(x)} = \lim_{x \rightarrow 0} \frac{\frac{2-2x}{(x^2+1)^2}}{f'(x)} = 2$$

$$4.11) x^2 \cdot \frac{dy}{dx} = -y \Rightarrow \frac{1}{x^2} dx = -\frac{1}{y} dy, \text{ 两边积分:}$$

$$-\frac{1}{x} + C_1 = -\ln y + C_2 \Rightarrow \ln y = \frac{1}{x} + C_3 \Rightarrow y = e^{\frac{1}{x}} \cdot C = Ce^{\frac{1}{x}}$$

$$12) x \cdot \frac{1}{\cos y} \cdot dx = -(x+1) dy \Rightarrow -\cos y dy = \frac{x}{x+1} dx = 1 \cdot dx - \frac{1}{x+1} dx$$

$$-\sin y + C_1 = x - \ln(x+1) + C_2 \Rightarrow \sin y = \ln(x+1) - x + C$$

$$(13) \quad \frac{1}{y} dy = \frac{x}{x^2+1} dx \Rightarrow \text{两边积分, 令 } u = x^2+1 \quad u' = 2x \quad du = 2x dx$$

$$\therefore \ln y + C_1 = \int \frac{\frac{1}{2} du}{u} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln u = \frac{1}{2} \ln(x^2+1) + C_2$$

$$\Rightarrow \ln y = \frac{1}{2} \ln(x^2+1) + C_3 \Rightarrow y = C(x^2+1)$$

$$(14) \quad y \cdot \frac{dy}{dx} + e^{y^2} \cdot e^{3x} = 0 \Rightarrow y \cdot \frac{dy}{dx} = -e^{y^2} \cdot e^{3x} \Rightarrow y e^{-y^2} dy = -e^{3x} dx,$$

$$\text{令 } u = -y^2 \quad du = -2y dy, \quad y dy = -\frac{1}{2} du \quad \therefore \frac{1}{2} e^u du = e^{3x} dx, \text{ 两边积分}$$

$$\frac{1}{2} e^u = \frac{1}{3} e^{3x} \Rightarrow 3e^u = 2e^{3x} + C \Rightarrow 3e^{-y^2} = 2e^{3x} + C$$

$$5.11) \quad \frac{dy}{dx} = \frac{y}{x} + \sec \frac{y}{x}, \quad \text{令 } u = \frac{y}{x}, \quad \frac{dy}{dx} = u + \sec u = u + x \frac{du}{dx} \Rightarrow \frac{1}{\sec u} du = \frac{1}{x} dx$$

$$\cos u \cdot du = \frac{1}{x} dx, \text{ 两边积分:}$$

$$\sin u + C_1 = \ln|x| + C_2 \Rightarrow \ln|x| = \sin \frac{y}{x} + C$$

$$(2) \quad (x+y) dx = (x-y) dy \Rightarrow \frac{dy}{dx} = \frac{x+y}{x-y} = \frac{1+\frac{y}{x}}{1-\frac{y}{x}} \quad \text{令 } u = \frac{y}{x}$$

$$\text{有 } x \frac{du}{dx} = \frac{1+u}{1-u} - u = \frac{1+u-u+u^2}{1-u} = \frac{u^2+1}{1-u} \Rightarrow \frac{1-u}{u^2+1} du = \frac{1}{x} dx = \frac{1}{u^2+1} du - \frac{u}{u^2+1} du, \text{ 两边积分}$$

$$\text{令 } u^2+1 = m \quad m' = 2u \Rightarrow dm = 2u du \quad \therefore \int \frac{u}{u^2+1} du = \int \frac{\frac{1}{2} dm}{m} = \frac{1}{2} \int \frac{1}{m} dm = \frac{1}{2} \ln m = \frac{1}{2} \ln(u^2+1)$$

$$\therefore \arctan u - \frac{1}{2} \ln(u^2+1) = \ln|x| + C \Rightarrow \arctan \frac{y}{x} - \frac{1}{2} \ln\left(\frac{y^2}{x^2} + 1\right) = \ln|x| + C$$

$$(13) \quad x \cdot \frac{dy}{dx} = y + x e^{\frac{y}{x}} \Rightarrow \frac{dy}{dx} = \frac{y}{x} + e^{\frac{y}{x}} \quad \text{令 } u = \frac{y}{x} \quad -u = m \quad dm = -du \quad e^m dm$$

$$x \frac{du}{dx} = u + e^u - u = e^u \Rightarrow \frac{1}{x} dx = e^{-u} du, \text{ 两边积分:}$$

$$\ln x + C_1 = e^{-u} + C_2 \Rightarrow \text{有 } \ln x = -e^{-\frac{y}{x}} + C \Rightarrow \ln x = -e^{-\frac{y}{x}} + C$$

$$6.11) y' = \frac{dy}{dx} = \left(\frac{2}{x+y}\right)^2. \quad \text{令 } u = x+y. \quad y = u-x \quad y' = \frac{dy}{dx} = \frac{du}{dx} - 1$$

$$\frac{du}{dx} - 1 = \frac{4}{u^2} \Rightarrow \frac{du}{dx} = \frac{4}{u^2} + 1 = \frac{4+u^2}{u^2}$$

$$\therefore \frac{u^2}{4+u^2} du = dx \quad \text{两边积分:}$$

$$\int \frac{u^2}{4+u^2} du = x + C_1, \quad \int \frac{u^2}{4+u^2} du = \int 1 - \frac{4}{4+u^2} du = u - 4 \int \frac{1}{4+u^2} du = u - 2 \arctan \frac{u}{2} + C_2$$

$$\therefore \text{由 } x + C_1 = (x+y) - 2 \arctan \frac{x+y}{2} + C_2 \Rightarrow y = 2 \arctan \frac{x+y}{2} + C$$

$$12) y' = \frac{2(x+2y)+3}{x+2y+1}. \quad \text{令 } u = x+2y. \quad y = \frac{u-x}{2}. \quad y' = \frac{1}{2} \cdot \frac{du}{dx} - \frac{1}{2} = \frac{2u+3}{u+1} \Rightarrow \frac{du}{dx} - 1 = \frac{4u+6}{u+1}$$

$$\frac{du}{dx} = \frac{4u+6+u+1}{u+1} = \frac{5u+7}{u+1} \Rightarrow dx = \frac{u+1}{5u+7} du \quad \text{两边积分}$$

$$\int 1 dx = \frac{1}{5} \int \frac{5u+5}{5u+7} du = \frac{1}{5} \int 1 - \frac{2}{5u+7} du = \frac{1}{5} u - \frac{2}{5} \int \frac{2}{5u+7} du \quad \text{令 } m = 5u+7 \quad dm = 5 du$$

$$\Rightarrow du = \frac{1}{5} dm \quad \therefore \int \frac{2}{5u+7} = \frac{2}{5} \int \frac{dm}{m} = \frac{2}{5} \ln m + C_2 = \frac{2}{5} \ln(5u+7) + C_2$$

$$\therefore x + C_1 = \frac{1}{5} u - \frac{2}{5} \ln(5u+7) + C_2 = \frac{1}{5} (x+2y) - \frac{2}{5} \ln(5x+10y+7) + C_2$$

$$\therefore 5x + C = (x+2y) - \frac{2}{5} \ln(5x+10y+7) \quad \therefore 25x + C = 5x+10y - 2 \ln|5x+10y+7|$$

$$\therefore 20x + C = 10y - 2 \ln|5x+10y+7|$$

$$13) \text{ 令 } u = 4x+y \quad y = u-4x. \quad \frac{dy}{dx} = y' = \frac{du}{dx} - 4 = \frac{1}{u(u+1)}$$

$$\frac{du}{dx} = \frac{1}{u^2+u} + 4 = \frac{4u^2+4u+1}{u^2+u} \Rightarrow dx = \frac{u^2+u}{4u^2+4u+1} du = \frac{\frac{1}{4}(4u^2+4u+1) - \frac{1}{4}}{4u^2+4u+1} du = \frac{1}{4} du - \frac{1}{4} \frac{1}{(2u+1)^2} du$$

$$\text{两边积分: } x + C_1 = \frac{1}{4} u - \frac{1}{4} \int \frac{1}{(2u+1)^2} du. \quad \text{令 } m = 2u+1 \quad dm = 2 du$$

$$\therefore \int \frac{1}{(2u+1)^2} du = \int \frac{\frac{1}{2} dm}{m^2} = \frac{1}{2} \int m^{-2} dm = -\frac{1}{2m} + C_3$$

$$\therefore x + C_1 = \frac{1}{4} x + \frac{1}{4} y + \frac{1}{8m} + C_2 \Rightarrow \frac{1}{4} y + \frac{1}{16u+8} + C = 0 \Rightarrow \frac{1}{4} y + \frac{1}{8} \cdot \frac{1}{2u+1} + C = 0$$

$$2u + \frac{1}{2u+1} + C = 0 \Rightarrow 2y + \frac{1}{8x+2y+1} + C = 0$$

$$7. (1) \frac{dy}{y^2+1} = dx \text{ 两边积分: } \int \frac{1}{1+y^2} dy = \int 1 \cdot dx$$

$$\text{解: } \arctan y = x + C \quad \text{又: } y(1) = 0 \quad \arctan 0 = 1 + C = 0 \Rightarrow C = -1$$

$$\therefore y = \tan(x-1)$$

$$(3) \frac{dy}{dx} = \frac{x-y+1}{x-y} \quad \text{令 } u = x-y \quad y = x-u \Rightarrow \frac{du}{dx} = 1 - \frac{du}{dx}$$

$$\therefore \frac{dy}{dx} = 1 - \frac{du}{dx} = \frac{1}{u} + 1 \Rightarrow -\frac{du}{dx} = \frac{1}{u} \Rightarrow -u du = dx \text{ 两边积分:}$$

$$\text{解: } -\frac{1}{2} u^2 = x + C \Rightarrow -\frac{1}{2} (x-y)^2 = x + C \quad y(0) = 1$$

$$\therefore -\frac{1}{2} (0-1)^2 = 0 + C \Rightarrow C = -\frac{1}{2}$$

$$\therefore \text{解: } -\frac{1}{2} (x-y)^2 = x - \frac{1}{2} \Rightarrow x = \frac{1}{2} - \frac{1}{2} (x-y)^2 \Rightarrow 2x = 1 - (x-y)^2$$

$$8. (1) y = C(x) e^{-\int P(x) dx}, \quad C(x) = \int e^x e^{\int P(x) dx} dx + C$$

$$\int P(x) dx = \int -3 dx = -3x$$

$$C(x) = \int e^{-2x} dx + C = -\frac{1}{2} e^{-2x} + C$$

$$\therefore y = (-\frac{1}{2} e^{-2x} + C) \cdot e^{3x} = C e^{3x} - \frac{1}{2} e^x$$

$$(3) y = C(x) e^{-\int (-2x) dx} \quad C(x) = \int x \cdot e^{\int (-2x) dx} dx$$

$$\int (-2x) dx = -x^2$$

$$\therefore C(x) = \int x e^{-x^2} dx \quad \text{令 } u = -x^2 \quad du = -2x dx \Rightarrow x dx = -\frac{1}{2} du$$

$$\therefore C(x) = -\frac{1}{2} \int e^u du = -\frac{1}{2} e^u + C = -\frac{1}{2} e^{-x^2} + C$$

$$y = (-\frac{1}{2} e^{-x^2} + C) e^{x^2} = C e^{x^2} - \frac{1}{2}$$

$$(4) y' + \cos x \cdot y = e^{-\sin x}$$

$$y = C(x) e^{-\int (\cos x) dx} \quad C(x) = \int e^{-\sin x} \cdot e^{\int (\cos x) dx} dx$$

$$\int (\cos x) dx = \sin x \quad \text{bzw.} \quad C(x) = \int e^{-\sin x} \cdot e^{\sin x} dx = \int 1 dx = x + C$$

$$y = (x + C) e^{-\sin x} = x e^{-\sin x} + C e^{-\sin x}$$

$$9.4) \quad y' + y = x + e^x$$

$$y = C(x) e^{-\int dx}, \quad C(x) = \int (x + e^x) \cdot e^{\int dx} dx$$

$$\int dx = x$$

$$\therefore C(x) = \int (x + e^x) e^x dx = \int x e^x + e^{2x} dx = (x-1) e^x + \frac{1}{2} e^{2x} + C$$

$$y = \left[(x-1) e^x + \frac{1}{2} e^{2x} + C \right] e^{-x} = (x-1) + \frac{1}{2} e^x + C e^{-x}$$

$$\text{w.} \quad y(0) = 0. \quad \text{f\u00fcr:} \quad 0 = (-1) + \frac{1}{2} + C \Rightarrow C = \frac{1}{2}$$

$$\therefore \text{f\u00fcr:} \quad 2y = 2x - 2 + e^x + e^{-x}$$

$$10.11) \quad y' - 3xy = xy^2 \Rightarrow \frac{y'}{y^2} - 3xy^{-1} = x. \quad \text{f\u00fcr } u = y^{-1} \quad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{du}{dx} = -y^{-2} \frac{dy}{dx} \Rightarrow \frac{du}{dx} = -y^2 \frac{du}{dx}$$

$$\frac{-du}{dx} - 3xu = x$$

$$\therefore -u' - 3xu = x \Rightarrow u' + 3xu = -x$$

$$\text{f\u00fcr } u = C(x) \cdot e^{-\int 3x dx} \quad C(x) = \int (x) e^{\int 3x dx} dx$$

$$\int 3x dx = \frac{3}{2} x^2 \quad \therefore C(x) = \int (x) e^{\frac{3}{2} x^2} dx \quad \text{f\u00fcr } m = \frac{3}{2} x^2 \Rightarrow dm = 3x dx$$

$$C(x) = \frac{1}{3} \int e^m dm = \frac{1}{3} e^m + C = \frac{1}{3} e^{\frac{3}{2} x^2} + C = -\frac{1}{3} e^{\frac{3}{2} x^2} + C_1$$

$$u = \frac{1}{y} = \left(\frac{1}{3}e^{\frac{3}{2}x^2} + C\right)e^{-\frac{3}{2}x^2} = \frac{1}{3} + Ce^{-\frac{3}{2}x^2}$$

$$\textcircled{2} \frac{dy}{dx} + \frac{2y}{x} = \frac{y^3}{x^2} \Rightarrow \frac{dy}{dx} \cdot y^{-3} + \frac{2}{x} \cdot y^{-2} = \frac{1}{x^2}, \text{ 令 } u = y^{-2} \quad du = -2y^{-3} dy$$

$$\frac{du}{dx} = -2y^{-3} \cdot \frac{dy}{dx} \quad \frac{dy}{dx} \cdot y^{-3} = -\frac{1}{2} \frac{du}{dx}$$

$$\therefore -\frac{1}{2} \frac{du}{dx} + \frac{2}{x} \cdot u = \frac{1}{x^2} \Rightarrow u' - \frac{4}{x}u = \frac{-2}{x^2}$$

$$u = C(x) \cdot e^{-\int(-\frac{4}{x})dx} \quad C(x) = \int(-\frac{2}{x^2})e^{\int(-\frac{4}{x})dx} dx$$

$$\int(-\frac{4}{x})dx = -4\ln x \quad C(x) = \int(-\frac{2}{x^2})e^{-4\ln x} dx = \int(-\frac{2}{x^2}) \cdot x^{-4} dx$$

$$= -2 \int x^{-6} dx = \frac{2}{5}x^{-5} + C$$

$$u = \left(\frac{2}{5}x^{-5} + C\right) \cdot x^4 = Cx^4 + \frac{2}{5}x^{-1} = \frac{1}{y^2}$$

$$\therefore \frac{1}{y^2} = \frac{2}{5x} + Cx^4$$

$$11. f(x) = \int_0^{2x} f\left(\frac{t}{2}\right) dt + \ln 2, \quad f'(x) = 2f(x)$$

$$\text{令 } f(x) = y, \text{ 有 } y' = 2y \Rightarrow \frac{dy}{dx} = 2y \Rightarrow \frac{1}{y} dy = 2dx \Rightarrow \ln|y| + C_1 = 2x + C_2$$

$$\therefore y = Ce^{2x}, \text{ 当 } x=0, f(0) = \ln 2$$

$$\therefore C = \ln 2, \therefore f(x) = e^{2x} \ln 2$$

$$12. \text{ 令 } t = ux, u = \frac{t}{x} \Rightarrow du = \frac{dt}{x}, \int_0^x f(t) dt = \frac{x}{2} f(x) + x$$

$$\text{两边求导: } f(x) = \frac{x}{2} f'(x) + \frac{1}{2} f(x) + 1, \quad \frac{1}{2} f(x) = \frac{x}{2} f'(x) + 1, \quad \text{令 } f(x) = y$$

$$\frac{1}{2}y = \frac{x}{2}y' + 1 \Rightarrow y = xy' + 2 \Rightarrow y' - \frac{y}{x} = -\frac{2}{x}$$

$$y = C(x) \cdot e^{-\int(-\frac{1}{x})dx} \quad C(x) = \int -\frac{2}{x} \cdot e^{\int(-\frac{1}{x})dx} dx$$

$$\int (-\frac{1}{x}) dx = -\ln|x| \quad C(x) = \int -\frac{2}{x} \cdot \frac{1}{x} dx = -2 \int \frac{1}{x^2} dx = 2 \cdot \frac{1}{x} = \frac{2}{x} + C$$

$$\therefore y = (\frac{2}{x} + C) \cdot x = 2 + Cx$$

13. 两边求导: $2y(x) \cdot \sqrt{1+y'^2(x)} = 2 + 2y(x) \cdot y'(x)$

$$\Rightarrow y(x) \cdot \sqrt{1+y'^2(x)} = 1 + y(x) \cdot y'(x) \Rightarrow y \sqrt{1+(y')^2} = 1 + y'y$$

两边平方: $y^2 \cdot [1+(y')^2] = 1 + (y')^2 y^2 + 2y'y$

$$\therefore y^2 = 1 + 2y'y \Rightarrow \frac{y^2-1}{2y} = \frac{dy}{dx} \Rightarrow dx = \frac{2y}{y^2-1} dy \text{ 两边积分}$$

$$\text{令 } m = y^2 - 1 \quad dm = 2y dy \quad \therefore \int \frac{2y}{y^2-1} dy = \int \frac{dm}{m} = \ln|m| + C_2$$

$$\therefore x + C_1 = \ln|y^2-1| + C_2 \Rightarrow \ln|y^2-1| = x + C$$

$$\Rightarrow y^2-1 = Ce^x \Rightarrow y^2 = Ce^x + 1$$

$$\Rightarrow y(x) = \sqrt{Ce^x + 1}$$

14. $\frac{dR}{dt} = -kR \Rightarrow -k dt = \frac{1}{R} dR$ 两边积分:

$$-kt + C = \ln|R| \Rightarrow R = Ce^{-kt}$$

$$C = R_0 \quad \frac{1}{2}R_0 = R_0 e^{-k \cdot 1600} \Rightarrow \frac{1}{2} = e^{-1600k}$$

$$\therefore 1600k = \ln 2 \Rightarrow k = \frac{\ln 2}{1600}$$

$$\therefore R = R_0 e^{-\frac{\ln 2}{1600} t}$$

$$\frac{1}{10}R_0 = R_0 e^{-\frac{\ln 2}{1600} t} \Rightarrow t = 5315. \therefore \text{再过 } 3715 \text{ 年}$$

$$17.11) \frac{dH}{dt} = k(H-T_0). \text{ 故 } \frac{dH}{dt} = k(H-20) \Rightarrow \frac{dH}{H-20} = k dt \Rightarrow \ln|H-20| + C_1 = kt + C_2$$

$$\therefore \ln|H-20| = kt + C \Rightarrow H-20 = Ce^{kt} \Rightarrow H = Ce^{kt} + 20, t=0, H=37$$

$$\therefore C=17$$

$$t=2, H=35 \text{ 有: } 35 = 17e^{2k} + 20 \Rightarrow \frac{15}{17} = e^{2k} \Rightarrow k = \frac{1}{2} \ln \frac{15}{17}$$

$$\therefore H = 17e^{\left(\frac{1}{2} \ln \frac{15}{17}\right)t} + 20$$

$$12) 32 = 17e^{\frac{1}{2} \ln \frac{15}{17} t} + 20 \Rightarrow e^{\left(\frac{1}{2} \ln \frac{15}{17}\right)t} = \frac{12}{17} \Rightarrow \ln \frac{12}{17} = \left(\frac{1}{2} \ln \frac{15}{17}\right)t \Rightarrow t = \frac{2 \ln \frac{12}{17}}{\ln \frac{15}{17}} = 5.57 \text{ h} = 5 \text{ h } 34 \text{ min}$$

$$\therefore 10:26 \text{ am 发电}$$

$$y' + p(x)y = Q(x)$$

$$18. \text{ 设电流为 } i. E = iR + 2 \frac{di}{dt} \Rightarrow 20 \sin t = 10i + 2 \frac{di}{dt} \Rightarrow \frac{di}{dt} + 5i = 10 \sin t$$

$$\text{有: } y = C(x) \cdot e^{-\int 5 dt} \quad C(x) = \int 10 \sin t \cdot e^{\int 5 dt} dt \quad \int 5 dt = 5t$$

$$\therefore C(x) = 10 \int \sin t \cdot e^{5t} dt \quad \text{令 } m = 5t, dm = 5 dt$$

$$\therefore C(x) = 10 \int \sin m \cdot e^m \cdot \frac{1}{5} dm = 2 \int \sin m e^m dm = e^m (\sin m - \cos m) = e^{5t} (\sin t - \cos t)$$

$$\therefore y = [e^{5t} (\sin t - \cos t) + C] \cdot e^{-5t} = \sin t - \cos t + Ce^{-5t} = i$$

$$19. \text{ 设室内 CO 浓度为 } x. V_2 = 0.05 \times 6\% = 0.003 \text{ m}^3/\text{min} \quad V_d = 0.06x \text{ m}^3/\text{min}$$

$$\frac{d(500x)}{dt} = 0.03 - 0.06x \Rightarrow 500 \frac{dx}{dt} = 0.03 - 0.06x = 0.01(3 - 6x) \Rightarrow \frac{0.01}{500} dt = \frac{dx}{3-6x}$$

$$\frac{0.01}{500} t = -\frac{1}{6} \ln(3-6x) + C \Rightarrow \frac{-0.06t}{500} = \ln(3-6x) + C \quad \frac{1}{50} t = 0, x = 0, C = -\ln 3$$

$$\therefore \frac{1}{50} x = 0.1\% \quad \ln(2.994) - \ln 3 = \frac{-0.06t}{500} \Rightarrow t = 168.3 \text{ min}$$

$$1.1) y' = 3x^2 - C \Rightarrow C = 3x^2 - y'$$

$$\therefore y = x^3 - (3x^2 - y')x = -2x^3 + xy'$$

$$\therefore y - xy' + 2x^3 = 0$$

$$2.1) xy' + y = y \ln xy \quad \text{令 } u = xy \quad y = \frac{u}{x} \quad u' = y + xy' \Rightarrow xy' = u' - y$$

$$\therefore xy' + y = u' = y \ln u \Rightarrow \frac{du}{dx} = \frac{u}{x} \ln u \Rightarrow \frac{1}{x} dx = \frac{1}{u \ln u} du$$

$$\text{令 } m = \ln u. \quad \text{则 } dm = \frac{1}{u} du \Rightarrow \frac{1}{m} dm = \frac{1}{x} dx \Rightarrow \ln|m| + C_1 = \ln|x| + C_2$$

$$\Rightarrow C_1 m = x \Rightarrow C_1 \ln u = x \Rightarrow C_1 \ln(xy) = x \Rightarrow \ln(xy) = \frac{x}{C_1} \Rightarrow xy = Ce^x$$

$$12) \frac{dx}{dy} = \frac{x^4 + y^2}{4x^3y} = \frac{x}{4y} + \frac{y}{4x^3}$$

$$13) y' = \frac{-2y}{y^2 - 6x} = \frac{2y}{6x - y^2} \Rightarrow \frac{dx}{dy} = \frac{6x - y^2}{2y} = \frac{3x}{1} - \frac{y}{2} \Rightarrow x' - \frac{3}{y}x = -\frac{y}{2}$$

$$\therefore x = C(x) \cdot e^{-\int -\frac{3}{y} dy} \quad C(x) = \int -\frac{y}{2} \cdot e^{\int -\frac{3}{y} dy} dy \quad y^{-2}$$

$$\int -\frac{3}{y} dy = \ln y^{-3} \quad e^{-3 \ln y} = \frac{1}{y^3} \Rightarrow C(x) = -\int \frac{y}{2} \frac{1}{y^3} dy = -\frac{1}{2} \int \frac{1}{y^2} dy = \frac{1}{2y} + C$$

$$x = \left(\frac{1}{2y} + C\right) \cdot (y^3) = Cy^3 + \frac{y^2}{2}$$

$$17) e^y - y' = \frac{2}{x} \Rightarrow y' = e^y - \frac{2}{x} = \frac{dy}{dx} \quad \text{令 } u = e^y \quad \therefore u' = y' e^y$$

$$3. \begin{cases} f'(-x) - x f'(x) = -x \\ f'(x) + x f'(-x) = x \end{cases} \Rightarrow f'(-x) = x f'(x) - x \Rightarrow f'(x) + x^2 f'(x) - x^2 = x$$

$$\Rightarrow \frac{x^2 + x}{1 + x^2} = f'(x) \Rightarrow f'(x) = 1 + \frac{x-1}{x^2+1}$$

$$f(x) = x + \int \frac{x}{x^2+1} dx - \int \frac{1}{x^2+1} dx = x + \frac{1}{2} \ln(x^2+1) - \arctan x + C$$

4. 对于微分方程 $y' + p(x)y = Q(x)$. 有: $y = C(x)e^{-\int p(x) dx}$ $C(x) = \int Q(x)e^{\int p(x) dx} dx$

$$\begin{cases} y_1' + p(x)y_1 = Q(x) \\ y_2' + p(x)y_2 = Q(x) \end{cases} \quad \text{同构方程: 令 } u = y_2 - y_1 \Rightarrow u' = y_2' - y_1'$$

$$\therefore \text{有 } u' + p(x)u = 0 \quad (\frac{2}{1} \text{ 边})$$

$$\text{若令 } m = y_2 - y_1, \text{ 则有: } m' + p(x)m = 0.$$

$$\therefore \frac{y_2 - y_1}{y_2 - y_1} = C$$

5. $y' + ky = f(x) \Rightarrow y = C(x)e^{-\int k dx}$, $C(x) = \int f(x) \cdot e^{\int k dx} dx$

$$e^{\int k dx} = e^{kx}. \quad C(x) = \int f(x)e^{kx} dx, \quad y = \int e^{kx} f(x) dx \cdot e^{kx}$$